

Sample Paper 6 — Answer Key (Level-2 Difficulty Step-Up)

Total: 60 marks (45×1 mark + 5×3 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **D) Shark** — Whale, Dolphin, and Bat are mammals; Shark is a fish.
2. **A) Music** — An author creates a book; a composer creates music.
3. **A) RCKPV** — The rule is "+2 letters": $P \rightarrow R, A \rightarrow C, I \rightarrow K, N \rightarrow P, T \rightarrow V$.
4. **D) 91** — Two patterns are interleaved: the odd-position terms go 2, 4, 8, 16 (doubling); the even-position terms go 100, 97, 94, ? (-3 each) — so the next even term is 91.
5. **A) Aman, Chetan, Bala, Esha, Dinesh** — Bala is in the middle (position 3). Aman is at the left end (position 1). Chetan sits between them (position 2). Esha sits between Bala and Dinesh, so Esha is position 4 and Dinesh is position 5.
6. **C) Back at the starting point** — North and South cancel (9 m each way); East and West cancel (7 m each way).
7. **B) Tuesday** — $26 - 5 = 21$ days later; 21 is exactly 3 weeks, so the day of the week repeats: Tuesday.
8. **D) 150** — $27 = 3 \times 3 \times 3$, $64 = 4 \times 4 \times 4$, $125 = 5 \times 5 \times 5$ (each a number multiplied by itself twice — a "cube number"); 150 does not fit this pattern.
9. **D) 720** — Each term is multiplied by one more than the last multiplier used ($\times 2, \times 3, \times 4, \times 5, \times 6$): $120 \times 6 = 720$.
10. **C) 9th** — Adding children at the left end pushes everyone's "position from the left" up by that many: $7 + 2 = 9$.
11. **D) 67** — 17, 34, and 51 are multiples of 17; 67 is not.
12. **A) RNCPGV** — The rule is "+2 letters": $P \rightarrow R, L \rightarrow N, A \rightarrow C, N \rightarrow P, E \rightarrow G, T \rightarrow V$.
13. **D) Cube** — Square, Rhombus, and Trapezium are flat (2-D) shapes; a Cube is a solid (3-D) shape.
14. **B) Team Y** — The second-highest team scored between Team X and Team Y, meaning Team Y scored more than that team. Since Team Z is highest overall and Team Y isn't the highest, Team Y must be the second-highest.
15. **D) 56** — The gaps between terms are 3, 6, 10, 15, 21 — each one 1, 2, 3, 4, 5 more than the last gap ($+3$, then $+4$ more to get $+6$, the gap itself increases by 1 more each time: 3, $6(+3)$, $10(+4)$, $15(+5)$, $21(+6)$, so the next gap is $+7$: $35+21=56$).

Section 2 — Mathematical Reasoning

16. **A) 7,44,443** — $4,56,789 + 2,87,654 = 7,44,443$.
17. **A) 4,43,211** — $9,00,000 - 4,56,789 = 4,43,211$.
18. **A) 95,046** — $6,789 \times 14 = 95,046$.
19. **A) 4,116** — $98,784 \div 24 = 4,116$ exactly.
20. **C) 24** — HCF of 72 ($2^3 \times 3^2$) and 96 ($2^5 \times 3$) is $2^3 \times 3 = 24$.
21. **B) 60** — Multiples of 15: 15,30,45,60... Multiples of 20: 20,40,60... First common one = 60.
22. **C) 7/9** — As decimals: $5/7 \approx 0.714$, $3/4 = 0.75$, $7/9 \approx 0.778$, $8/11 \approx 0.727$. The greatest is $7/9$.
23. **B) 4 km 250 m** — $2450 \text{ m} + 1800 \text{ m} = 4250 \text{ m} = 4 \text{ km } 250 \text{ m}$.
24. **C) 432 sq m** — Area = $24 \times 18 = 432 \text{ sq m}$.
25. **C) 12 cm** — A regular pentagon has 5 equal sides: $60 \div 5 = 12 \text{ cm}$.
26. **C) 30%** — Profit = $1560 - 1200 = ₹360$; profit % = $360/1200 \times 100 = 30\%$.
27. **B) 12.5%** — If principal P doubles, the interest earned equals P itself. $SI = (P \times R \times T) \div 100 \rightarrow P = (P \times R \times 8) \div 100 \rightarrow R = 100 \div 8 = 12.5\%$.
28. **C) ₹140** — Total parts = $5 + 6 + 7 = 18$; 1 part = $360 \div 18 = 20$; largest share = $7 \times 20 = ₹140$.
29. **A) 1/8** — $0.125 = 125/1000$, which simplifies to $1/8$.
30. **B) 9 kg 50 g** — $4500 \text{ g} + 2750 \text{ g} + 1800 \text{ g} = 9050 \text{ g} = 9 \text{ kg } 50 \text{ g}$.
31. **C) 9,000** — $2.5 \text{ hours} \times 3600 \text{ seconds/hour} = 9,000 \text{ seconds}$.
32. **B) 69 cm** — An equilateral triangle has 3 equal sides: $3 \times 23 = 69 \text{ cm}$.
33. **C) 840 sq cm** — Area = $35 \times 24 = 840 \text{ sq cm}$.
34. **C) 9** — A triangular prism has 2 triangular faces (3 edges each) plus 3 connecting edges = 9 edges.
35. **C) 9** — Number of diagonals = $n(n-3)/2 = 6 \times 3/2 = 9$.

Section 3 — Everyday Mathematics

36. **C) ₹23,000** — $1,84,000 \div 8 = 23,000$.
37. **B) 1,200** — Rate = $2,160 \div 9 = 240 \text{ bottles/hour}$; in 5 hours = $240 \times 5 = 1,200$.
38. **C) 18 min** — $3,600 \div 200 = 18 \text{ minutes}$.
39. **B) 4 h 15 min** — Total time 7:15 a.m. to 11:50 a.m. = 4 h 35 min; subtracting the 20-minute stop gives 4 h 15 min of actual travel.
40. **B) 1 L 250 mL** — $3250 \text{ mL} + 1750 \text{ mL} = 5000 \text{ mL}$; $5000 \div 4 = 1250 \text{ mL per bottle}$.
41. **C) ₹230** — $8 \times ₹32 = ₹256$; $₹256 - ₹26 \text{ discount} = ₹230$.
42. **C) 750 L** — $1/8 \times 1,200 = 150 \text{ L used per day}$; over 3 days = 450 L used; $1,200 - 450 = 750 \text{ L remaining}$.
43. **B) 66 kg** — $22500 \text{ g} + 18750 \text{ g} + 15250 \text{ g} + 9500 \text{ g} = 66,000 \text{ g} = 66 \text{ kg}$.
44. **C) 40** — $1,520 \div 38 = 40 \text{ rows}$.

45. **C) ₹640** — Sold for $80 \times ₹48 = ₹3,840$; profit = $₹3,840 - ₹3,200 = ₹640$.

Section 4 — Achievers Section (3 marks each)

46. **C) 22** — Working backwards: $(108 + 24) \div 6 = 132 \div 6 = 22$.

47. **C) 28** — Let the first number be x . Then second = $2x$, third = $2x+10$. Sum: $x + 2x + (2x+10) = 150$
 $\rightarrow 5x+10 = 150 \rightarrow x = 28$.

48. **B) ₹1,750** — After spending $2/7$ on books, $5/7$ of the money remains. After spending $1/5$ of that on stationery, $4/5$ of $5/7 = 4/7$ of the original remains. So $4/7$ of the total = ₹1,000 $\rightarrow$ total = ₹1,750.

49. **D) 208** — Girls = $(2 \times 100) + 8 = 208$. Check: $100 + 208 = 308$. ✓

50. **C) 34** — Each term is the sum of the previous two: $13 + 21 = 34$.

Scoring guide (out of 60, Level-2-difficulty paper): 32+ here is a very strong result — it's harder than Papers 1–5 by design, so the usual 42+ Level-1 benchmark doesn't directly apply. 22–31 is solid for a first attempt at this difficulty. Below 22 — that's expected the first time through; the value here is in reviewing *why* each missed question was harder (an extra step, a bigger number, two ideas combined) rather than the raw score.
